Large AI Model-Enabled Generative Semantic Communications for Image Transmission

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Agenda

- Background & Motivation
- Proposed Generative Semantic Communication System
- Model Lightweight and Fine-Tuning
- Simulation Results
- Conclusion & Future Work

Background & Motivation

Challenges in 6G and Image Transmission.

The Promise and Limitations of Semantic Communications.

The Challenge: Bridging 6G Demands and Transmission Reality

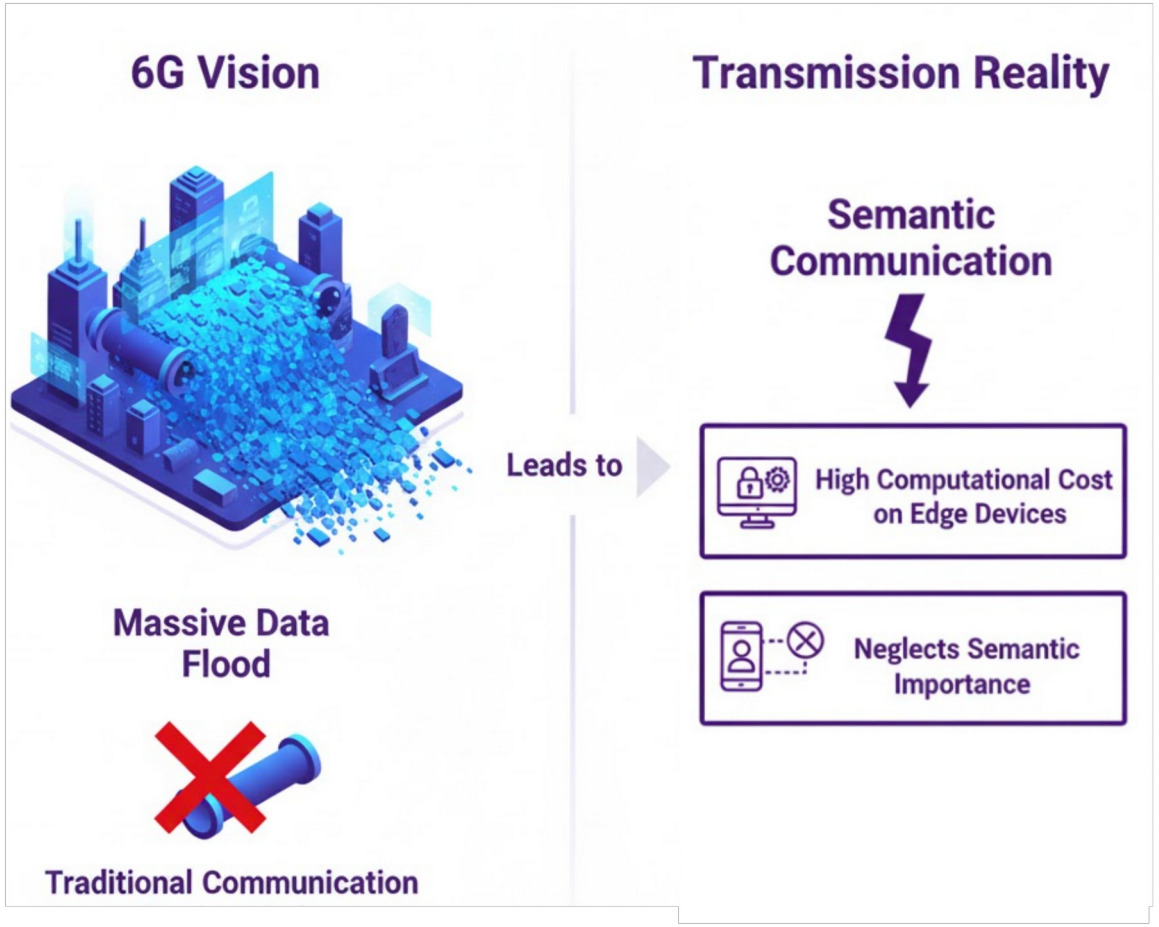
Questions

- **6G Vision:**

Ultra-high data rates, high-quality multi-modal communications

- **The Bottleneck:**

Traditional bit-oriented communication struggles with massive data from images/videos, leading to huge bandwidth consumption.



The Gap

- **Neglects Semantic Importance:** Treats all image regions equally, potentially degrading key object quality.
- **High Computational Cost:** Generative AI models are powerful but too resource-intensive for edge devices.

Our Solution: Semantic-Aware Generative Transmission

Core Idea

- Don't transmit the whole image. Instead, differentiate semantic importance.
- **Transmit Key Regions as Image Features.**
- **Compress Non-Key Regions into Text Prompts.**
- Leverage a generative model at the receiver to reconstruct the full image.

Main Contributions

- **A Novel Generative Framework:** Achieves fine-grained semantic representation by segmenting the image.
- **An Efficient Lightweight Strategy:** Employs QLoRA to make large models deployable on edge devices.
- **Superior Performance:** Outperforms traditional and deep learning methods, especially in low Signal-to-Noise Ratio (SNR) conditions.

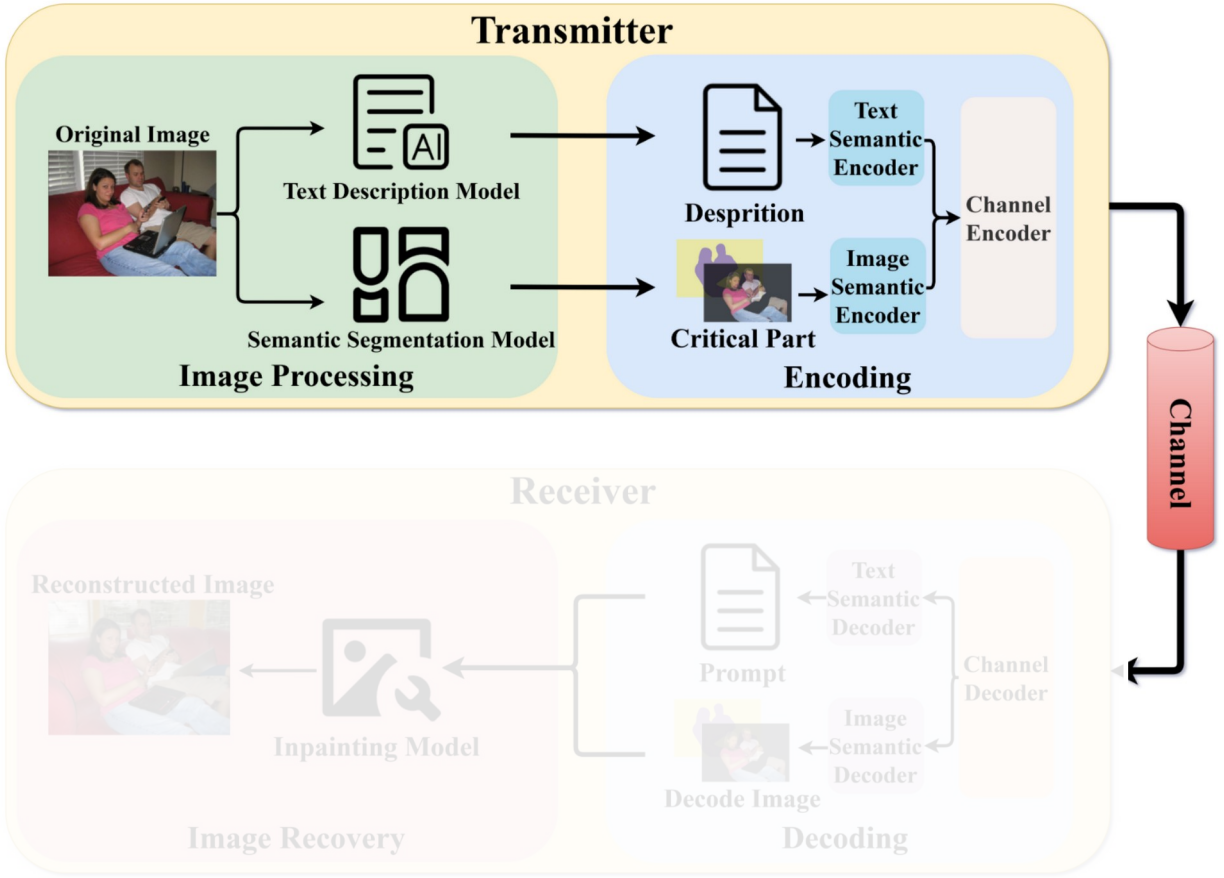
Proposed Generative Semantic Communication System

System Architecture and Core Idea.

Proposed Generative Semantic Communication System

- **Transmitter Side:**
  - **Input Image** : Start with the source image.
  - **Semantic Segmentation:** The image is divided into key regions and non-key regions.

$$Y_I = \mathcal{S}_\alpha(X) \in \mathbb{R}^m, \quad Y_T = \mathcal{D}_\psi(X) \in \mathbb{R}^n,$$



Proposed Generative Semantic Communication System

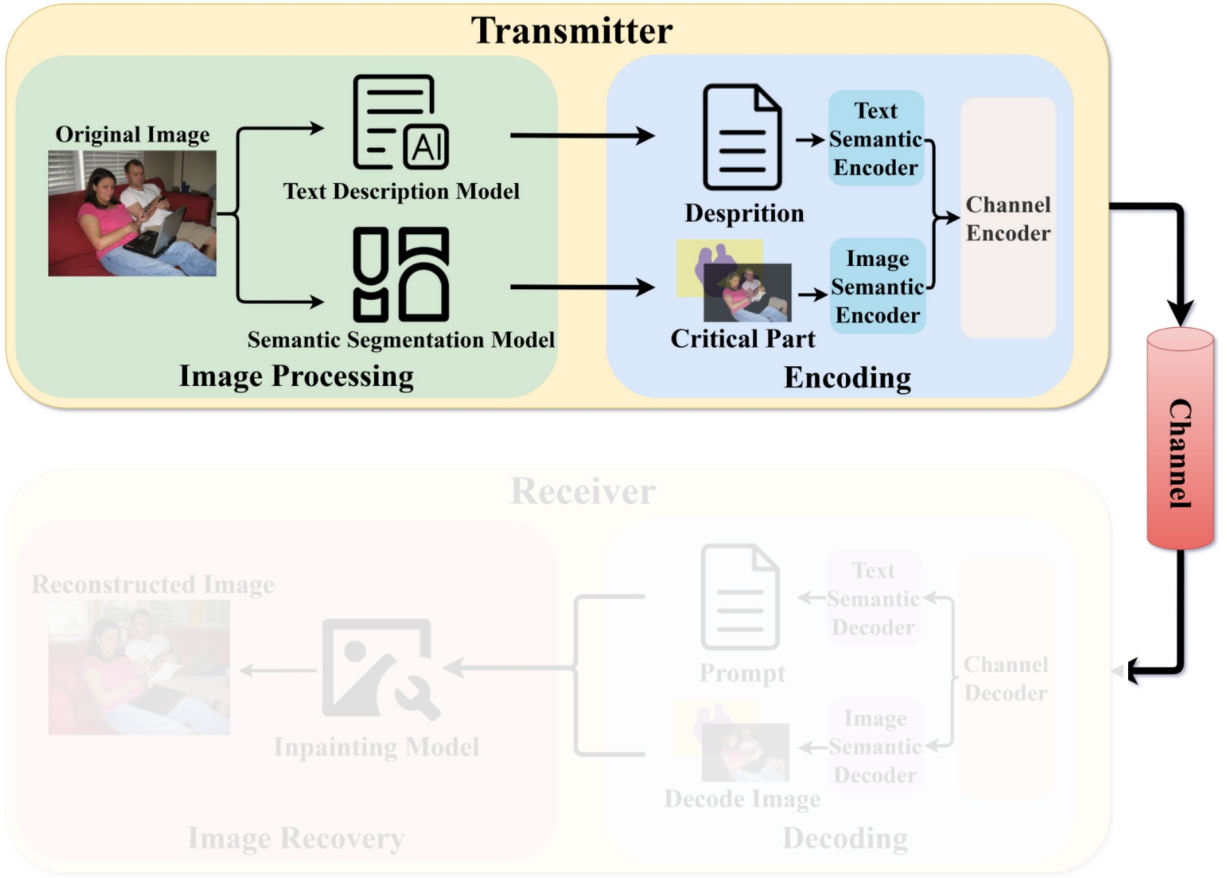
Transmitter Side:

- **Dual Encoding:** Key regions are encoded by an Image Semantic Encoder. Non-key regions are converted to Text Prompts by an image-to-text model.

$$Y' = \mathcal{E}_\theta(Y_I) \oplus \mathcal{E}_\phi(Y_T) \in \mathbb{R}^k,$$

- **Concatenation & Transmission:** Both encoded features are combined and sent over the wireless channel.

$$Y'' = hY' + \mathbf{n},$$



Proposed Generative Semantic Communication System

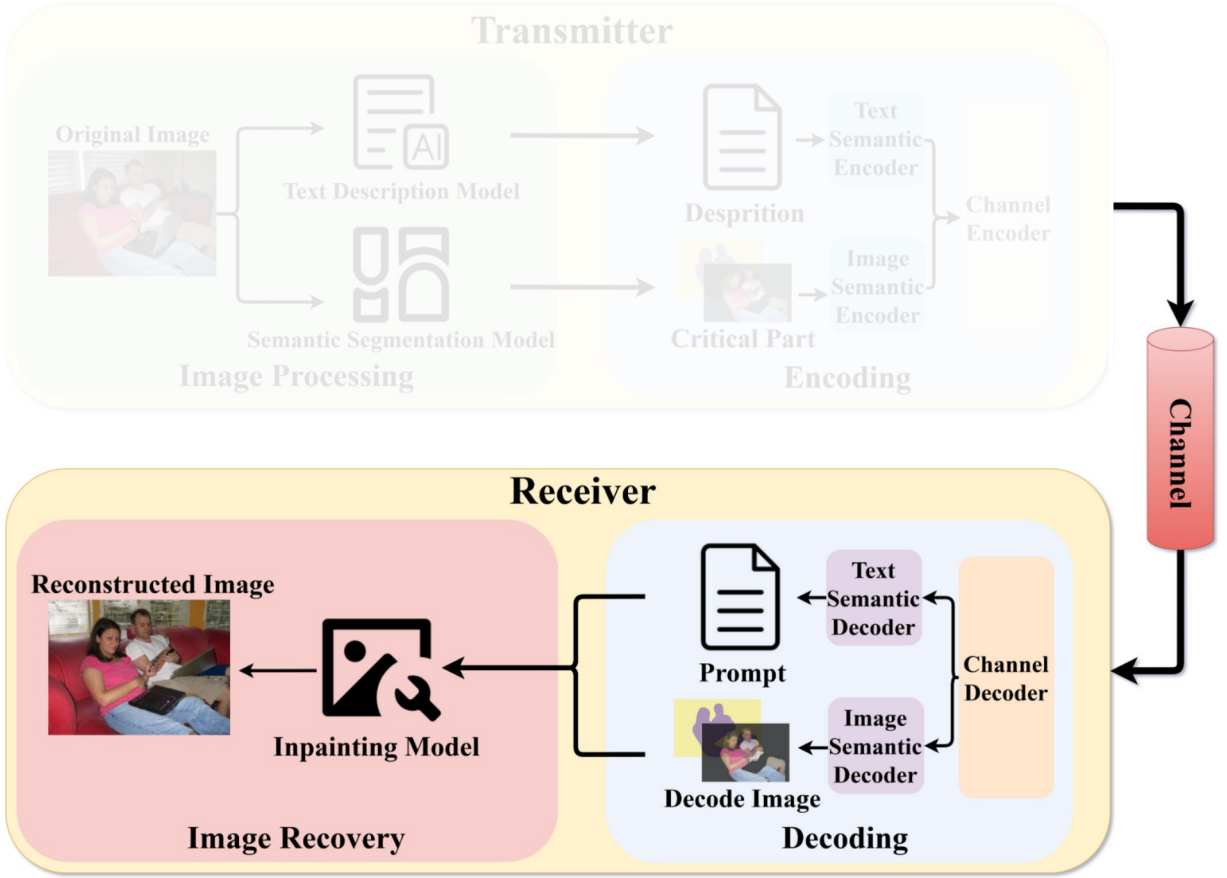
Receiver Side:

- **Decoding:** The received signal is split back into image features and text prompts .

$$\hat{Y}_I = \mathcal{D}_\theta(Y''_{1:k_1}) \text{ and } \hat{Y}_T = \mathcal{D}_\phi(Y''_{k_1+1:k})$$

- **Guided Generation:** A generative model uses both the image features and text prompts to reconstruct the final high-fidelity image .

$$dX_t = \underbrace{\left[f_\xi(X_t, t, \hat{Y}_I, \hat{Y}_T)\right]}_{\text{Drift term}} dt + \underbrace{g(t)d\bar{w}_t}_{\text{Diffusion term}} ,$$



Model Lightweight and Fine-Tuning

Making Large Models Practical for Edge Devices.

Challenge: Deploying Large Models on the Edge

- Generative models like BLIP-2 or Monkey are powerful but huge.

TABLE I
COMPARISON OF DIFFERENT MODELS

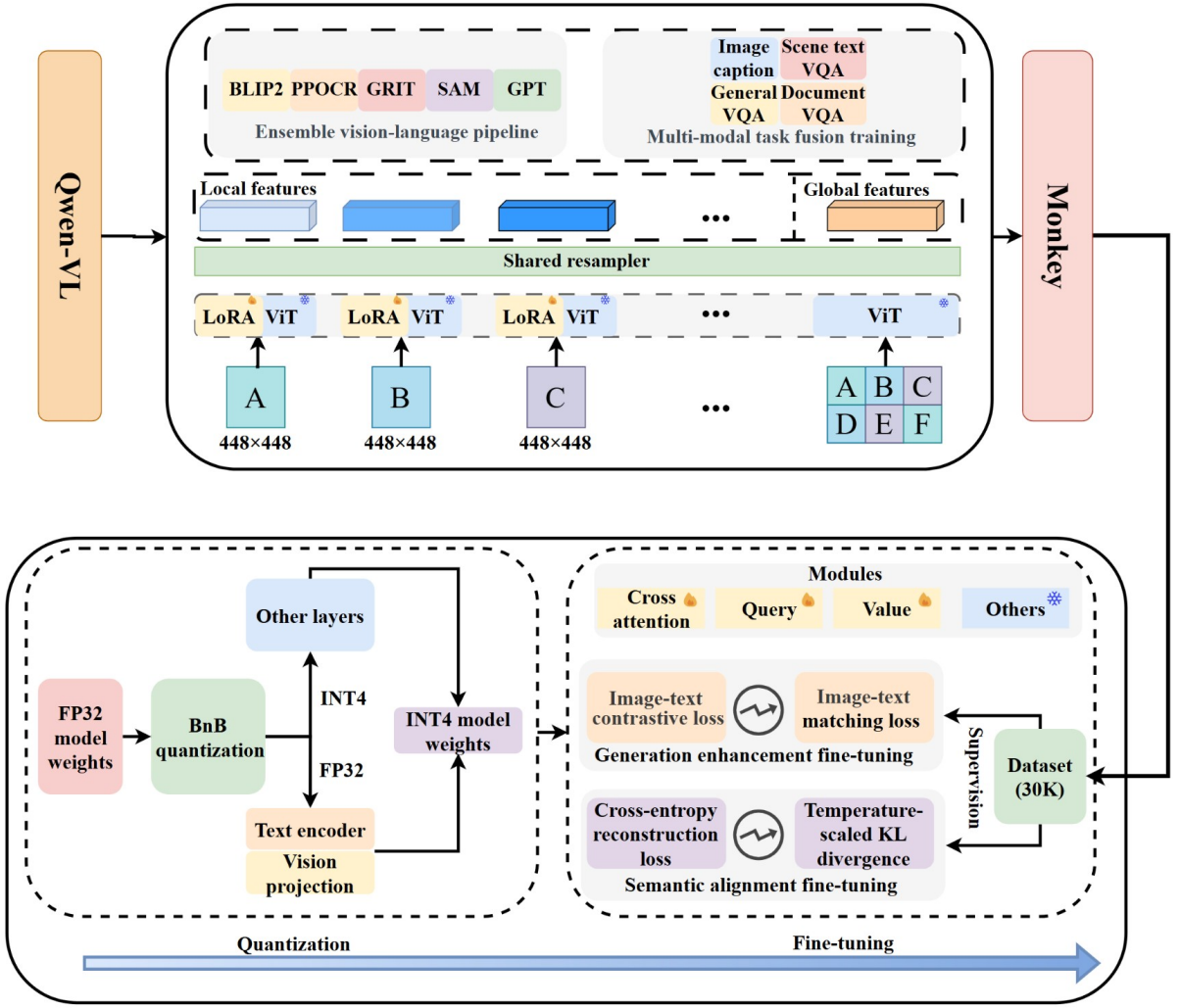
Models	Parameters (M)	Memory (GB)
Monkey	9708.05	18.09
BLIP	247.41	0.92
Proposed	**204.95**	**0.35**

- **A Three-Stage Optimization Strategy:**
  - **Model Quantization:**
    Use mixed-precision (INT4 + FP32) to drastically reduce
    memory.
  - **Parameter-Efficient Fine-Tuning (PEFT):**
    Use QLoRA to fine-tune the quantized model,
    recovering performance.
  - **Task-Oriented Training:**
    Fine-tune specifically for the image
    transmission task.

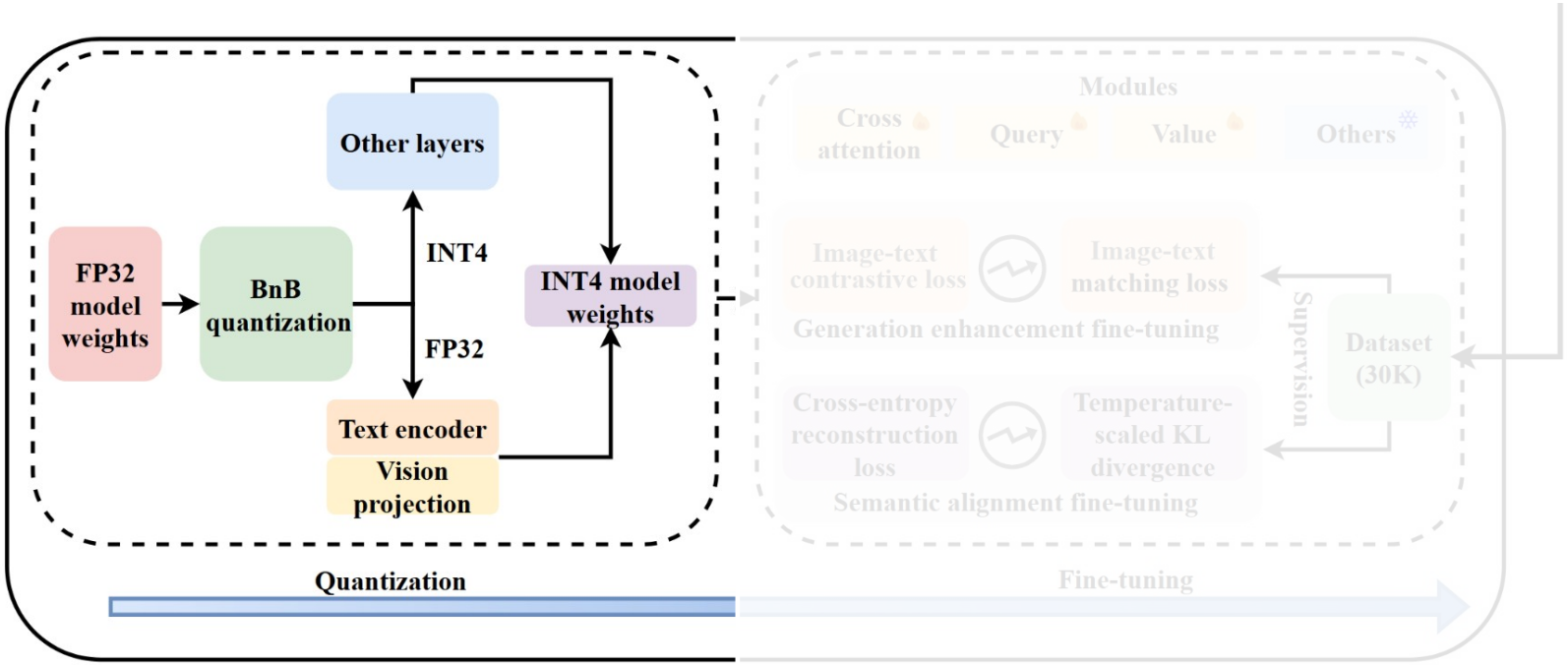
Lightweighting: Quantization and QLoRA Fine-Tuning

- To address the deployment challenges on resource-constrained edge devices, we propose a joint optimization framework integrating **Mixed-Precision Quantization** and **QLoRA Fine-Tuning**.

$$\Delta \mathbf{W}^* = \underset{\Delta \mathbf{W}}{\operatorname{argmin}} \mathcal{L}_{\text{total}} \left(f(x; \underbrace{\mathbf{W}_{\text{INT4}}}_{\text{Frozen}} + \underbrace{\Delta \mathbf{W}}_{\text{Trainable}}) \right)$$



Lightweighting: Quantization and QLoRA Fine-Tuning

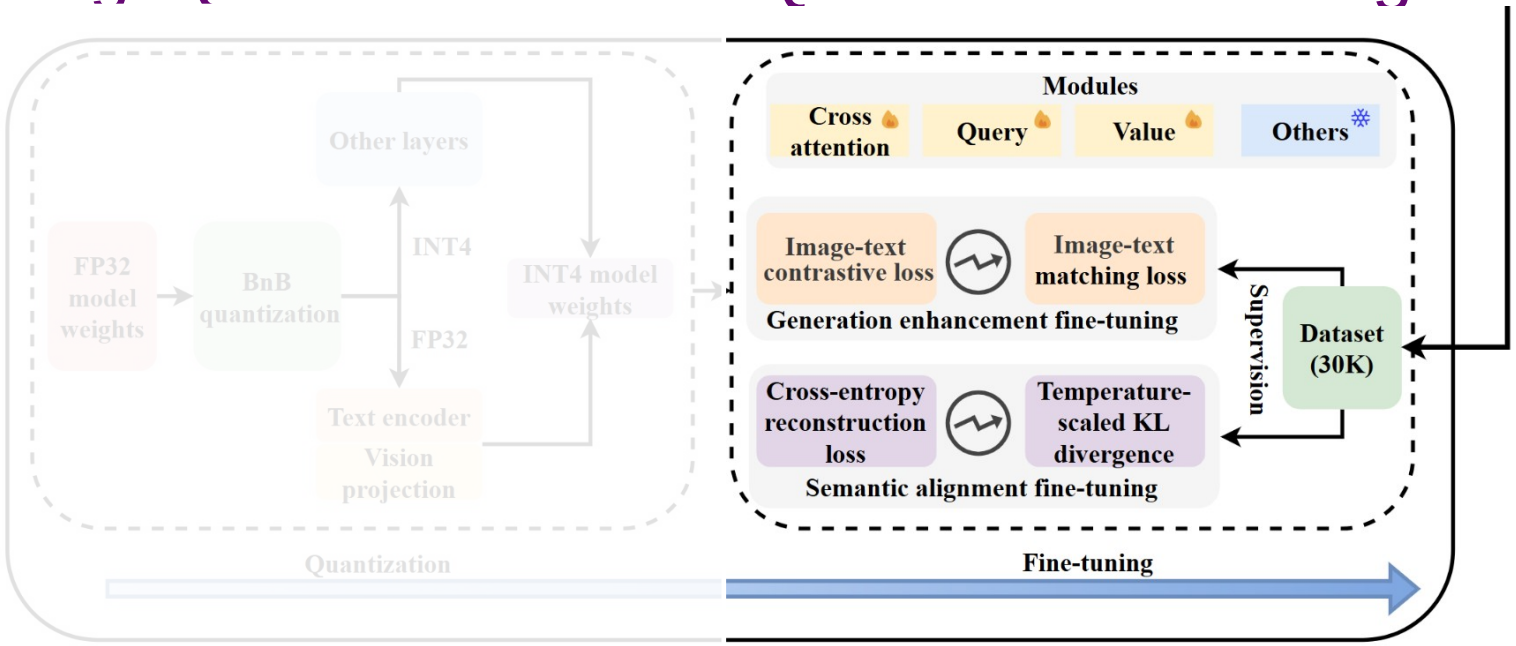


Part 1: Mixed-Precision Quantization

- We keep critical layers in full precision to preserve semantic fidelity, while quantizing the rest to reduce the model size

$$\mathbf{W}_{\text{BLIP-INT4}} = \begin{cases} \mathcal{Q}_{\text{BAB}}(\mathbf{W}_{\text{qua}}) & \text{quantized weights} \\ \mathbf{W}_{\text{crit}} & \text{original precision} \end{cases} \tag{7}$$

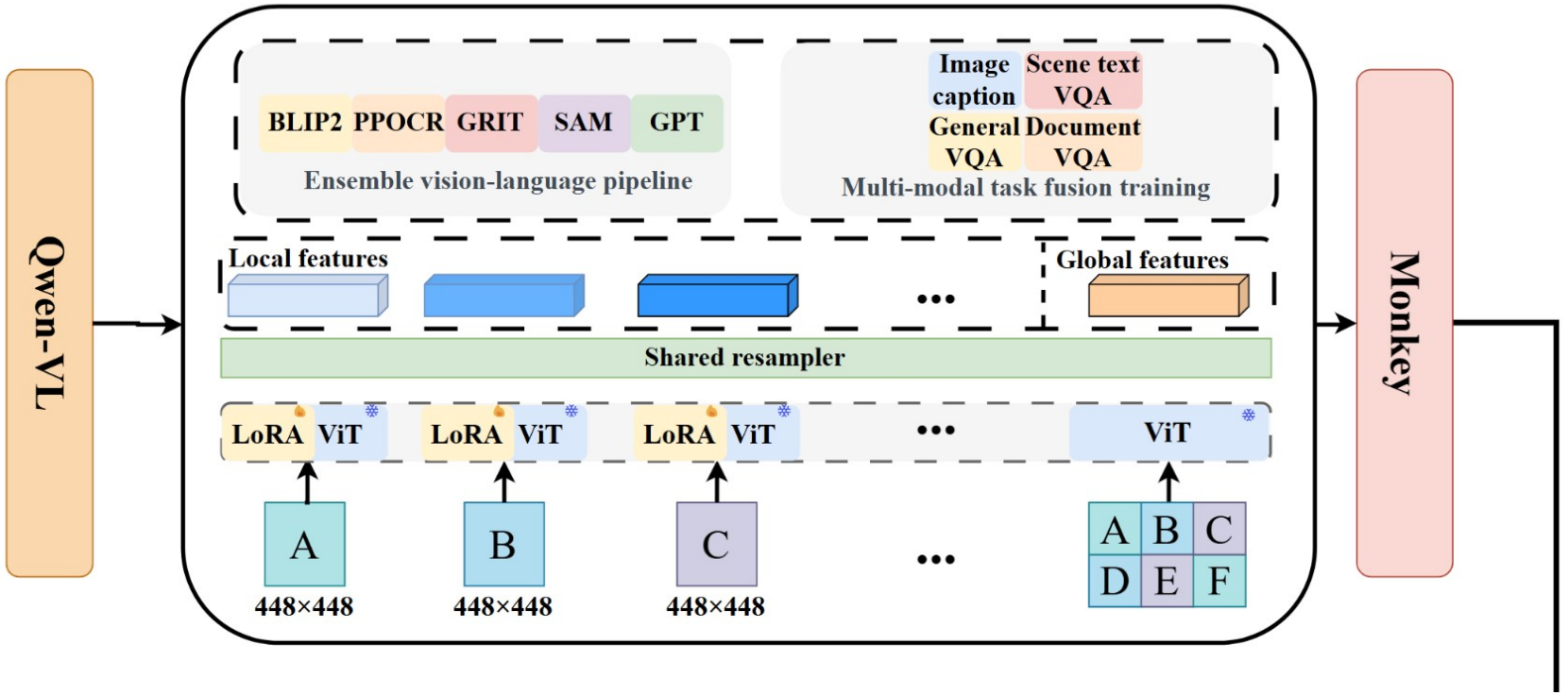
Lightweighting: Quantization and QLoRA Fine-Tuning



- **Part 2: QLoRA Fine-Tuning**
  - Instead of training the entire model, we only train small, low-rank adapters. This is highly efficient and works on quantized models.

$$\mathcal{L}_{\text{align}} = \lambda \mathcal{L}_{\text{ITC}} + (1-\lambda)\mathcal{L}_{\text{ITM}}, \quad \lambda \in [0,1],$$
$$\mathcal{L}_{\text{enhance}} = \beta \mathcal{L}_{\text{KL}} + (1-\beta)\mathcal{L}_{\text{CE}}, \quad \beta \in [0,1],$$

Lightweighting: Quantization and QLoRA Fine-Tuning



- **Fine-Tuning Source**
  - High-quality descriptions are generated using a well-established model (Monkey) and are used for the aforementioned fine-tuning after manual correction

Simulation Results

Performance Evaluation and Comparison.

Experimental Setup

- **Datasets:**
  - PIPA dataset (900 images) + Kodak24.
  - Foregrounds marked as "critical regions".
- **Baselines for Comparison:**
  - **Traditional:** BPG codec + LDPC channel coding.
  - **Deep Learning-based:** DeepJSCC .
  - **Ablation Studies:** DirectlyGen (only text), AllEdge (text + segmentation masks).
- **Evaluation Metrics:**
  - **Perceptual Quality:** LPIPS (↓), FID (↓), VSI (↑).
  - **Fidelity:** PSNR (↑), SSIM (↑), MSE (↓).

Original images

BPG-LDPC

proposed scheme







Compressed bits =5.05Kb

Compressed ratio = 0.89

Reduced overhead = 11%







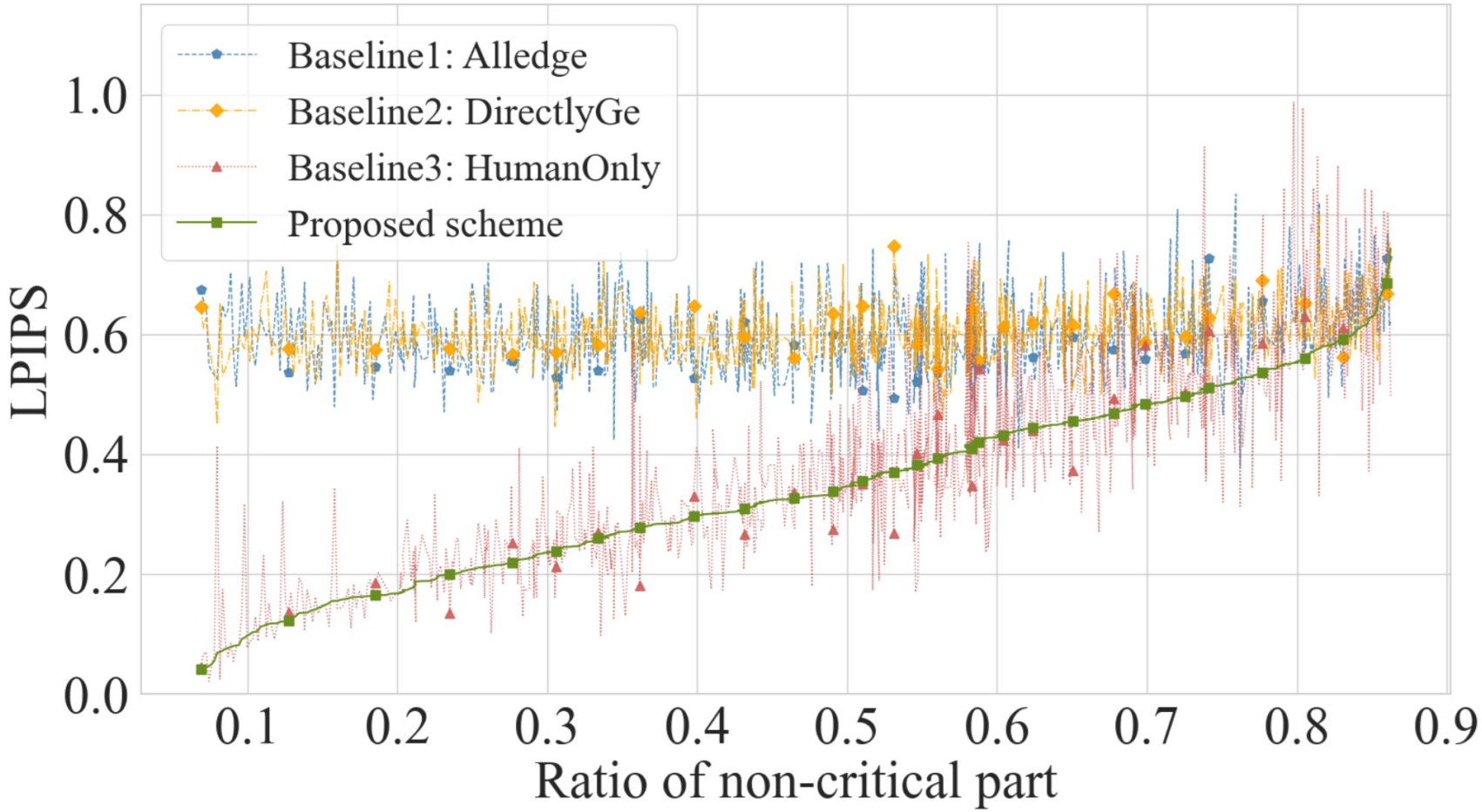
Compressed bits =10.05Kb

Compressed ratio = 0.88

Reduced overhead = 12%

Superior Visual Quality with Less Overhead

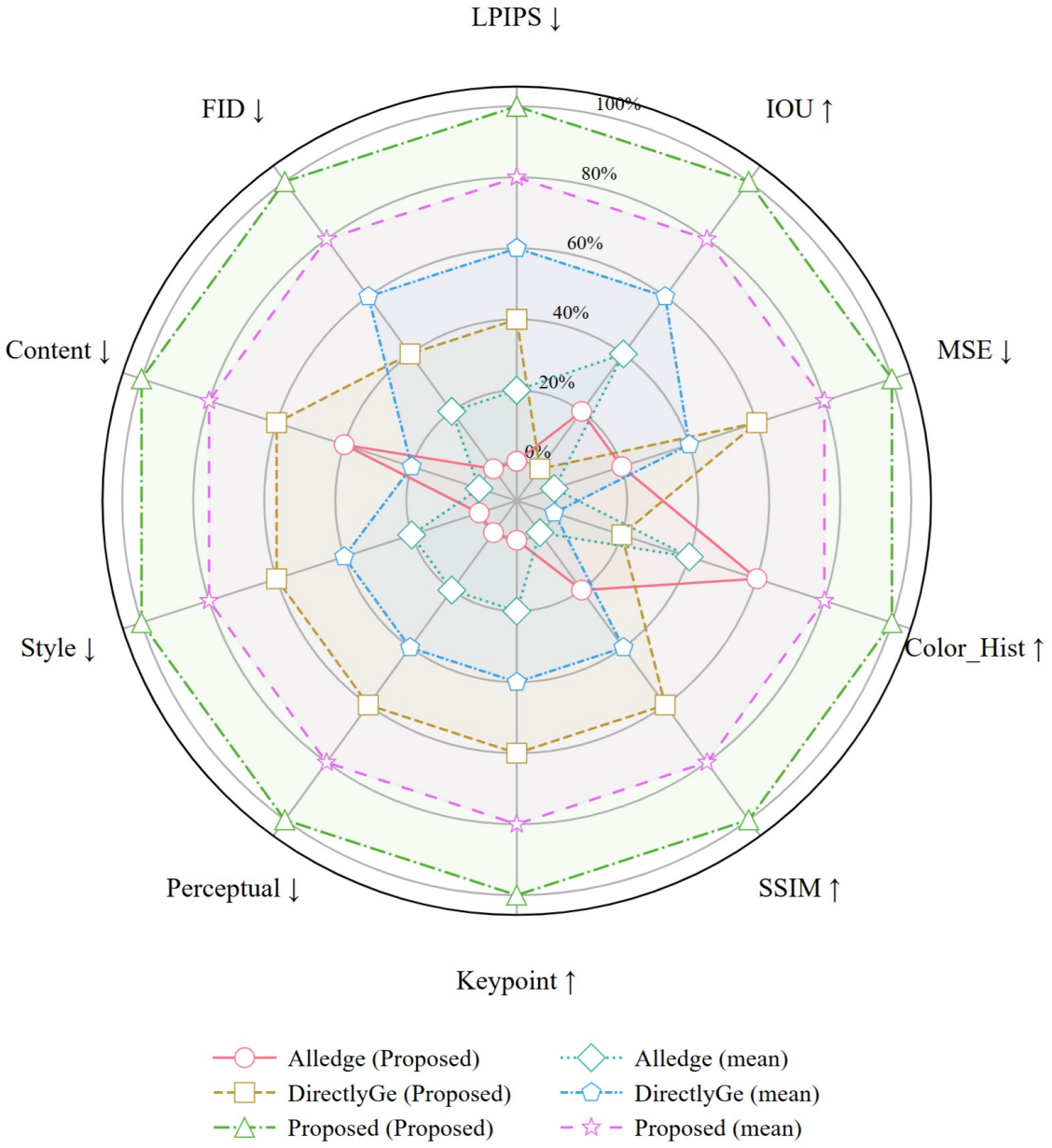
Our system not only reconstructs images with high fidelity but also significantly reduces the amount of data that needs to be transmitted.



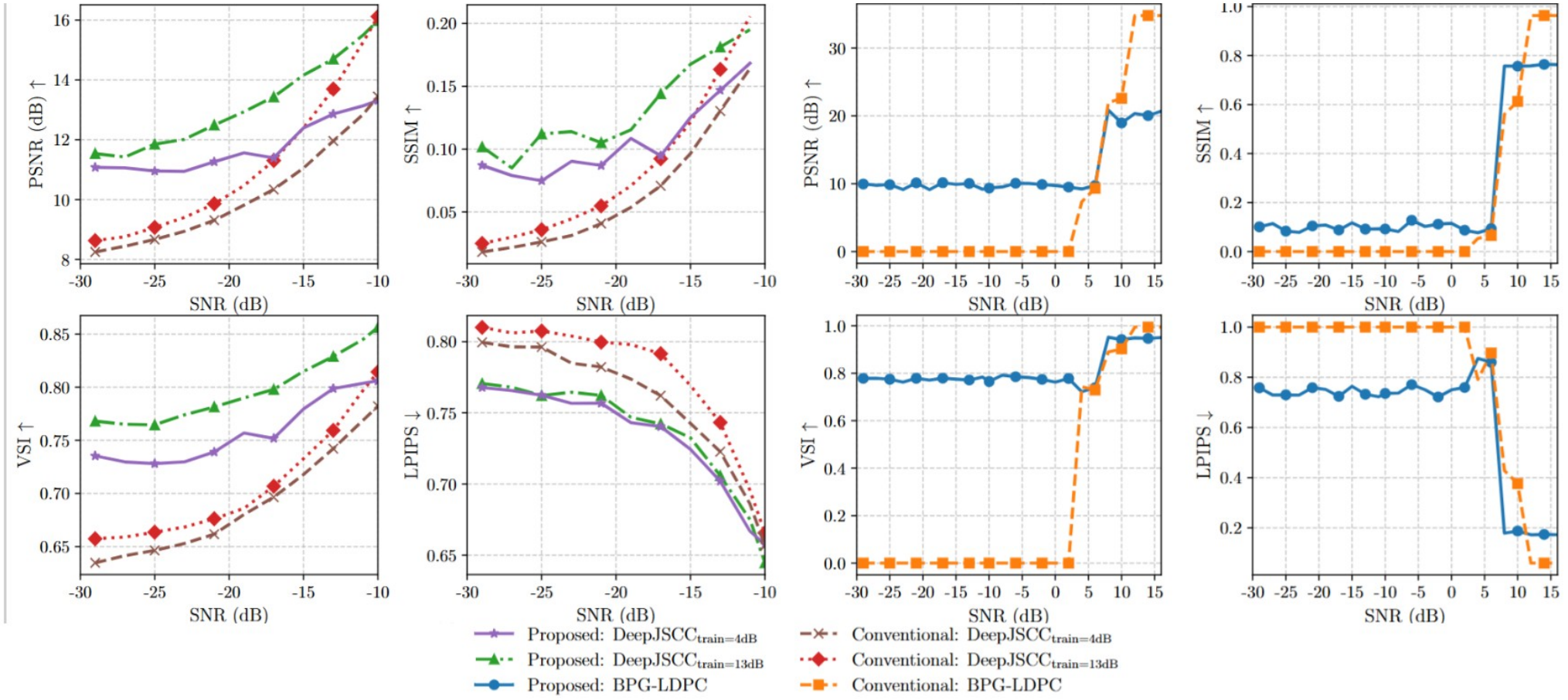
Ablation Study: Prompts are Crucial for Stability

This shows that text prompts act as a vital semantic anchor, guiding the generative model to produce consistent and high-quality results.

Comprehensive Evaluation: A Clear Winner



- Across a wide range of 10 different metrics, our proposed scheme consistently outperforms other methods, achieving the best overall performance.



Robustness Against Channel Noise (Low SNR Performance)

Our generative approach demonstrates remarkable resilience to channel noise. The learned semantic priors in the generative model help to 'imagine' the missing details, making it far more robust than conventional systems in harsh channel conditions.

Key Findings from Experiments

- **1. Higher Efficiency:** Achieved significant overhead reduction (e.g., 11-12%) while maintaining high visual quality.
- **2. Superior Quality:** Consistently ranked #1 across 10 diverse metrics, proving all-around excellence.
- **3. Exceptional Robustness:** Maintained significantly better performance under low SNR conditions compared to strong baselines like DeepJSCC.

Conclusion

Conclusion

- **We proposed a novel generative semantic communication system for image transmission.**
  - It intelligently separates key and non-key information, encoding the latter as text prompts to improve efficiency and bypass classical rate-distortion limits.
- **We designed an effective lightweighting strategy.**
  - Using quantization and QLoRA, we reduced the model's memory footprint by over 98% (vs. Monkey), making it practical for edge deployment.
- **The system demonstrated superior performance and robustness.**
  - It excels in visual quality, efficiency, and especially resilience to cha

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Thanks For Listening !
Q&A

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